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APPLIED RESEARCH

Long-Range Bluetooth Low Energy Detection via Drone-Based LoRa Relays for Search and Rescue Operations in Complex Terrain

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ABSTRACT Search and Rescue (SAR) missions in remote environments are time-critical and resource-intensive, and their effectiveness is often constrained by terrain, weather, and limited communications infrastructure. Bluetooth-based tracking offers a distinct advantage over other RF modalities due to its ubiquity in personal devices such as smartphones, smartwatches, and commercial trackers (e.g., AirTags), which continuously broadcast advertising packets that can be leveraged for localization. This work demonstrates the feasibility of using BLE advertisements as opportunistic signals for remote sensing in drone-based SAR missions to increase the efficiency of locating missing persons. A drone is used as a passive RF remote sensing node equipped with a Nordic nRF52840 scanner and different antenna configurations. To characterize the sensor's response and detection probability under varying propagation conditions extensive field tests across diverse locations in Spain, including slot canyons with no Global Navigation Satellite System (GNSS) or cellular coverage, dense forests, and high-mountain terrain have been conducted. The proposed system detected commercial BLE devices at slant ranges close to 500 m from altitudes of up to 100 m. To overcome the lack of cellular coverage in remote areas and identify the devices, a LoRa-based data bridge has been implemented. This bridge encapsulates and forwards captured BLE payloads to a ground control station, where they are re-emitted to trigger real-time updates on crowdsourced networks (e.g., Apple's Find My), effectively confirming the location of isolated victims to SAR command centers.

INDEX TERMS Antennas, Bluetooth, drones, LoRa, search and rescue (SAR), remote sensing, find my network.

I. INTRODUCTION

Search and rescue operations in remote and complex environments, such as high mountains, dense forests, and slot canyons are high-stakes missions where time is the most critical factor. Statistics highlight a persistent need for efficient rescue methodologies, as the probability of survival for a missing person decreases significantly after the first "golden" 24 hours [1]. For example, between 2004 and 2014, the U.S. National Parks alone required 46,609 SAR operations, incurring an approximate cost of $51.4 million [2]. In Spain, the Guardia Civil's mountain units had to organize rescues for 161 lost individuals in 2020, 207 in 2021, and 210 in 2022 [3], [4], [5].

The probability of detecting a missing person is often reduced by vast search areas and difficult environmental conditions, such as dense forests, rugged terrain, and adverse weather (e.g., fog, heavy rain, or snowfall). SAR efforts are typically carried out using ground teams and, in some cases, aerial support.

Ground-based SAR teams enable rapid deployment, direct terrain accessibility, and flexible planning, but they face challenges such as operator fatigue, limited situational awareness, and restricted sensor coverage. Aerial detection

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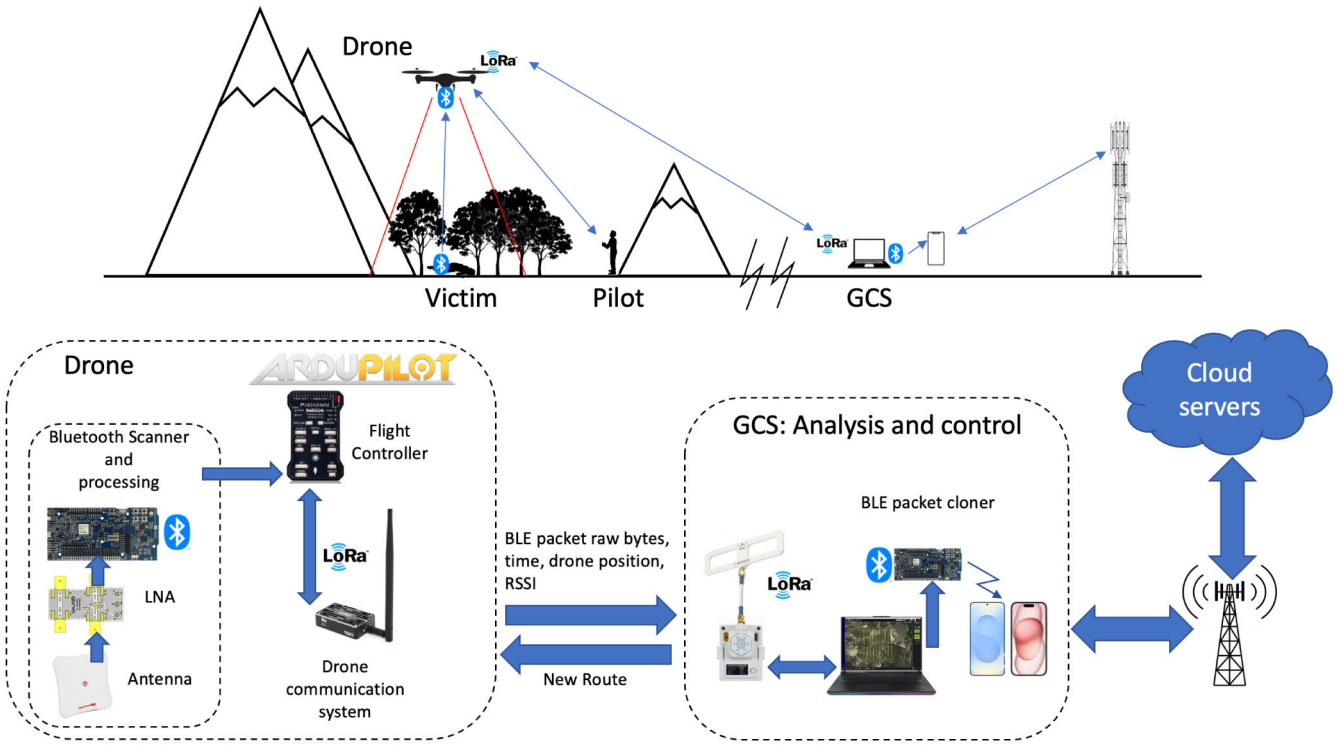


FIGURE 1. System architecture.

systems, on the other hand, offer several advantages over ground teams. They are unaffected by terrain irregularities, enable systematic search patterns, cover large areas quickly, and provide an overhead perspective that facilitates sensor-based detection. However, aerial searches are more susceptible to adverse weather conditions such as strong winds, heavy rain, or snow, and obstacles like trees and cliffs can obstruct direct visual detection. Traditionally, helicopters have been the primary tool for aerial SAR operations due to their ability to cover vast distances and reach remote areas. However, in recent years, Uncrewed Aerial Vehicles (UAVs) [6], commonly known as drones, have emerged as a complementary or alternative technology. Drones offer significant advantages in terms of cost, maneuverability, and rapid deployment, with recent research exploring multi-UAV cruising schemes and robust air-to-air channel models to guarantee swarm safety and continuous monitoring during critical operations [7].

Modern mountain SAR operations leverage technology to improve efficiency and effectiveness. Helicopters and UAVs equipped with thermal imaging enhance search capabilities in remote and snow-covered areas [8], satellite and Global Positioning System (GPS) tracking provide advanced location-based services to facilitate quicker victim identification, Artificial Intelligence (AI) and machine learning optimize search patterns by analyzing past missions and environmental conditions [9], avalanche transceivers and RECCO reflectors assist in locating buried victims [10], and communication networks such as satellite phones and real-time data transmission improve coordination among rescue teams. In this context, collaborative air-ground frameworks leveraging alpine networks and probabilistic search techniques have also been proposed to optimize operations in hostile mountain environments [11]. These technological tools reduce operational time, lower risks for SAR personnel, and increase the probability of successful rescues in extreme alpine conditions.

This paper presents a multi-technology relay SAR system, illustrated conceptually in Figure 1, which incorporates a BLE sniffer mounted on a drone and serves as an additional detection mechanism, enhancing SAR success rates. The research evaluates various antenna configurations, flight altitudes, speeds, and the feasibility of identifying specific devices associated with missing individuals when integrated with global crowdsourced location networks, such as Apple's "Find My" (a collaborative network to track lost items) by re-emitting captured packets at a connected base station. A key innovation of this study is the real-world evaluation of different configurations in different scenarios, demonstrating that commercial Bluetooth devices can be detected from altitudes up to 100 meters with coverage radius close to 500 meters. This capability enables the search of vast areas in a short time period.

Within the context of RF remote sensing, taking profit of the existing Bluetooth Low Energy (BLE) advertisements represents a shift towards opportunistic sensing, unlike traditional active sensing (e.g., Radar) or thermal imaging, which may be hindered by canopy cover or high equipment

costs. However, this modality presents significant challenges in terms of signal characterization, such as multi-path fading, shadowing by terrain features, and the non-stationary nature of the airborne sensor platform.

Despite the ubiquity of BLE-enabled devices, a significant technological gap remains in wilderness SAR operations. First, there is a generalized perception of BLE as a short-range technology limited to a few meters, which has hindered its adoption as a long-range remote sensing modality. Second, modern privacy standards in BLE utilize randomized MAC address rotation, making it nearly impossible for rescue teams to distinguish a victim's device from other incidental RF sources using only local sniffing. Previous research has either focused on very short-range detection (5–15 m) [12], [13] or required subjects to carry specific pre-registered beacons or equipment [14]. Our work addresses these limitations by transforming the UAV into a high-sensitivity aerial relay that extends detection to hundreds of meters and bridges captured packets to global crowdsourced networks such as Apple's Find My [15]. This allows the system to validate the identity of a specific device despite data anonymization, closing the loop between opportunistic RF sensing and actionable victim identification in communication-deprived environments.

Thus the following contributions are made: (1) an aerial passive RF remote sensing node using a BLE scanning system enhanced with a Low Noise Amplifier (LNA) on a drone; (2) a comprehensive experimental evaluation of this sensor across peri-urban forest, high-mountain, and slot-canyon environments under realistic regulatory constraints; and (3) an end-to-end BLE-to-LoRa-to-cloud relay that integrates commercial trackers with crowdsourced location networks such as Apple's Find My in communication-deprived environments.

The remainder of this paper is organized as follows. After reviewing the state of the art in modern SAR systems, section III explores the current landscape of BLE technology and the technical characteristics of crowdsourced search networks. Section IV details the proposed system architecture, focusing on the Bluetooth-LoRa bridge and the custom antenna design. The next section describes the experimental setup, including the target device configurations and the three field campaign scenarios. Section VI presents the results and a comparative discussion of the hardware performance. Finally, the conclusion summarizes the key findings and their implications for future SAR operations.

II. RELATED WORK

Recent systematic reviews of autonomous SAR operations emphasize a transition towards modular, five-layered architectures (comprising perception, navigation, and control layers) to improve scalability in dynamic disaster environments [16]. Within the perception layer, several detection technologies are currently employed in modern SAR operations to enhance the efficiency and accuracy of locating individuals. Image-based technologies, such as thermal imaging cameras and multispectral sensors, are typically deployed via drones or helicopters to identify heat signatures under low-visibility conditions [17]. A comprehensive state-of-the-art review on human detection from UAV aerial images can be found in [18], which extensively categorizes and evaluates different machine learning models, training datasets, and onboard hardware constraints in SAR scenarios. Radio Frequency (RF)-based approaches, including Bluetooth and LoRa, enable the tracking of signals emitted by personal devices, whereas mobile phone detection systems employ network triangulation and International Mobile Subscriber Identity (IMSI) catchers to determine an individual's position [19]. The RECCO system, a passive reflector technology extensively used in avalanche rescue operations, allows rescuers to identify victims buried beneath snow by detecting reflected radar signals.

Aerial SAR platforms employing visual detection techniques provide wide-area coverage and long-range identification capabilities through high-resolution zoom optics. However, UAV-based detection systems face operational challenges associated with the transmission of high-resolution imagery to ground control stations, where either human operators or AI algorithms [20] perform real-time analysis. This requires high-bandwidth communication links and reliable line-of-sight conditions.

RF detection systems play a critical role in SAR missions by facilitating the localization of distress signals transmitted by emergency beacons, mobile devices, or dedicated transponders. Such systems offer several advantages, including high detection sensitivity over extended distances, real-time tracking, and robust performance under adverse meteorological conditions that may degrade optical or infrared detection. Nonetheless, these systems are subject to interference from other RF sources and environmental noise, challenges in discriminating between genuine distress transmissions and false signals, and requirements for specialized equipment and trained personnel. Moreover, RF signal propagation is often affected by terrain features, potentially resulting in coverage gaps in mountainous environments.

Table 1 provides a comparative overview of several current SAR detection technologies summarizing their main characteristics, advantages, and limitations as analyzed in multiple studies discussed in this section.

One limitation of image-based systems is their dependence on high-speed communication links for data transmission. For instance, [21] reported a video transmission rate of 11.2 Mbps over a 5 km link using 2.4 GHz directional antennas, albeit under strict line-of-sight conditions.

Thermal and infrared imaging systems provide valuable complementary data for target detection. As demonstrated in [8], a thermal-imaging-based tracking system mounted on a UAV operating between 40 m and 60 m altitude enabled effective object detection and multi-target tracking during SAR missions. To optimize exploration efficiency in wide-area searches and minimize path redundancy, modern multi-drone frameworks have integrated shared thermal confidence maps and autonomous coordination algorithms [23]. Additionaly, more recent studies proposed a bio-inspired

TABLE 1. Comparison of current SAR detection technologies.

Technology	Estimated cost	Detection range	Required bandwidth	Advantages	Limitations	References
Image sensors	Low–Medium	Line-of-sight	Very high	Cost-effective, widely available	Limited by illumination and visibility conditions	[21] (2021) [22] (2026)
Thermal imaging	High	Up to 1 km	Very high	Effective in low visibility, detects heat sources	Performance affected by environmental occlusions such as vegetation or terrain	[8] (2024) [23] (2025)
Multispectral sensors	High	Several kilometers	Very high	Enhanced detection under diverse conditions	Expensive, requires specialized image processing	[17] (2024)
Bluetooth detection	Low	Hundreds of meters	Low	Detects consumer devices such as smartphones, headsets, and wearables	Limited by signal strength, duty cycle, and device battery life	[13] (2021) [12] (2022)
LoRa RF detection	Medium	Up to 10 km	Low	Long-range, low power consumption	Requires pre-registered devices	[24] (2020) [25] (2022) [26] (2024)
Mobile phone detection	Medium–High	Several kilometers (via network triangulation)	Low	Exploits existing telecom infrastructure	Dependent on cellular network coverage	[19] (2024)
RECCO system	Low	20–100 m	Low	Reliable in avalanche scenarios	Requires the user to carry a RECCO reflector	[14] (2024)

dual-altitude UAV swarm framework that utilizes both high-altitude thermal scanning for hotspot detection followed by low-altitude YOLO (You Only Look Once) visual refinement [22]. However, their performance can still be degraded by physical obstructions such as rocks, vegetation, and environmental temperature gradients.

RF technologies are broadly used in SAR applications. Notable examples include commercial systems such as ARVA, an avalanche victim detector standardized under ETSI EN 300 718-1 [27], and the RECCO system [14]. Although both were initially designed for avalanche rescue, their underlying principles can be adapted for broader SAR deployments. The RECCO system comprises two components: an active detector operated by rescuers and a passive reflector embedded in the subject's equipment. The detector emits a directed radar signal, which is reflected by the RECCO tag, thereby enabling pinpoint localization. The RECCO helicopter-mounted detector is capable of scanning approximately 1 km^2 in about six minutes from an altitude of 100 m. Despite being integrated into products from over 150 brands (such as jackets, helmets, and backpacks), not all outdoor users carry reflectors, limiting its applicability in generic SAR operations.

Recent studies have also investigated the use of Low-Power Wide-Area Network (LPWAN) technologies, such as LoRa, for SAR applications. LoRa-based localization has demonstrated promising results due to its long-range communication and low energy consumption. Experiments have shown that LoRa systems can transmit vital information, including health and location data, achieving ranges of up to 8.4 km under line-of-sight conditions [24]. However, a major limitation of pure LoRa tracking systems, even when presenting high channel quality and reliability under complex environments like avalanches [26], is that LoRa is not as ubiquitously integrated into everyday consumer electronic devices as Bluetooth, thereby requiring subjects to carry specific pre-registered beacons. On the other hand, LoRa has also been proposed as a long-range communication infrastructure between autonomous systems and base stations in SAR missions to handle real-time data exchange without relying on existing infrastructure [25], which conceptualizes our approach of deploying LoRa for the bridge between the airborne node and the ground station.

Despite the widespread availability of mobile phones, significant portions of rural and mountainous areas remain outside cellular network coverage. This limitation restricts the use of mobile applications for geolocation and prevents device-tracking services from providing real-time positional data, as these can only return the last known location within coverage.

Bluetooth-based detection has also been investigated for SAR purposes. In [12], an experimental BLE-based detection system mounted on a mobile ground platform demonstrated reliable identification of nearby devices within a 15 m range, even under challenging conditions. Since BLE is integrated into a wide range of consumer devices, including smartphones, smartwatches, headphones, and electronic tags, it represents a viable low-cost alternative for short-range detection. Moreover, commercial device-tracking ecosystems, such as Apple's *Find My* and Google's *Find My Device*, rely on BLE beacons to transmit encrypted location data to cloud infrastructures.

A drone-based SAR platform utilizing BLE detection was presented in [13], demonstrating the ability to survey a 3,600 m^2 area in less than five minutes. The system, evaluated in an indoor environment, achieved detection ranges of up to 5 m at low cost.

Despite the ubiquity of BLE devices, current research reveals a significant technological gap in wilderness SAR: the short-range nature of Bluetooth and the lack of cellular

backhaul render standard devices invisible to global tracking networks in remote areas. Our work addresses this gap by transforming the UAV into a high-sensitivity aerial relay. By integrating specialized RF front-ends to extend the detection radius and bridging this data through LoRa links, we enable the integration of isolated victims into crowdsourced ecosystems like Apple's Find My [15]. The following sections detail the hardware selection and the architectural design implemented to overcome these barriers.

III. BLE AND CROWDSOURCED SEARCH NETWORKS

Bluetooth is one of the most widespread wireless technologies, with annual shipments growing from 4.1 to 5 billion units in the last five years and a forecast of 7.5 billion by 2028 [28]. Operating in the 2.4 GHz Industrial, Scientific and Medical (ISM) band, BLE was introduced in 2010 (Bluetooth 4.0) to address the need for low-power, short-range communication in Internet of Things (IoT) and wearable sectors.

A. BLE PHY AND ADVERTISING MECHANISM

BLE divides the spectrum into 40 channels (2 MHz each). Unlike Classic Bluetooth, which focuses on continuous streaming, BLE is optimized for intermittent data transmission using a Gaussian Frequency-Shift Keying (GFSK) modulation.

Advertising is the primary way for devices to broadcast their presence. BLE uses three dedicated advertising channels to minimize interference from Wi-Fi: Channel 37 (2402 MHz), 38 (2426 MHz), and 39 (2480 MHz). Devices broadcast small packets (payloads up to 31 bytes) on these channels at intervals typically ranging from 20 ms to 10.24 seconds.

B. SIGNAL CHARACTERISTICS AND FCC PARAMETERS

The effectiveness of airborne detection depends on the device's Equivalent Isotropic Radiated Power (EIRP). Regulatory bodies like ETSI (EN 300 328) limit EIRP to 10 dBm, while FCC Part 15.247 allows for higher limits under specific conditions. However, typical consumer devices operate with lower transmitted power. Based on official FCC filings [29], we can characterize some usual devices as follows:

- **Smartphones**: Apple iPhone models (e.g., iPhone 15/16) [30] typically transmit around 9 dBm.
- **Trackers:** Apple AirTags [31] (based on nRF52832) transmit at 4.5 dBm with an antenna gain of −3.2 dBi.
- **Wearables:** Garmin devices (Instinct 2s [32], Forerunner series [33]) show EIRP values between −4 dBm and 3.12 dBm. Polar Vantage V3 [34] reports EIRP around −0.5 dBm.
- **Audio devices:** Apple AirPods Pro transmit at 10.04 dBm with −2.8 dBi gain [35], while Sony WF-1000XM5 [36] report approximately 8.3 dBm with −3 dBi gain.
- **IoT/New form factors:** The Galaxy Ring [37] exhibits a lower output of 3.33 dBm with a significantly reduced gain of −12.53 dBi due to its extremely small form factor.



FIGURE 2. Hexsoon EDU-650 taking off with full payload: sniffer, LNA and ImmersionRC antenna.

Thus, typical BLE EIRP values range from −4 dBm to about 10 dBm where, in most cases, antenna gains are below 0 dBi because these devices use compact, omnidirectional antennas with low radiation efficiency.

C. CROWDSOURCED NETWORKS: FIND MY AND FIND MY DEVICE

Apple and Google leverage BLE, asymmetric encryption, and a vast network of collaborative devices to track lost items. The process is decentralized and privacy-focused with the following steps:

1) **Broadcast:** The lost device broadcasts an encrypted, rotating public key via BLE advertising packets.
2) **Detection:** Any nearby finder device (third-party smartphone) captures the packet, retrieves its own location (GPS/Wi-Fi), and encrypts this location using the owner's public key.
3) **Relay:** The finder device anonymously transmits this encrypted bundle to the cloud.
4) **Decryption:** Only the owner, possessing the corresponding private key, can decrypt and view the location in the application.

In wilderness SAR, the absence of "finder" devices prevents this loop from closing. Our system acts as an airborne "finder" node, bridging the gap by relaying these packets via LoRa to an internet-connected gateway.

IV. MATERIALS AND METHODS

The methodology of this study follows a multi-stage approach aimed at validating the technical feasibility of the system architecture previously introduced in Figure 1. First, we describe the selection and integration of the aerial platforms and the specialized RF hardware, focusing on sensitivity enhancement through low-noise amplification. Then, we detail a comparative analysis of the three antenna geometries studied to optimize the search area. Finally, we present the integration of the Ground Control Station (GCS) and the software-based bridge designed to verify connectivity with global crowdsourced location networks.

A. AERIAL PLATFORMS

For the field tests, two different drones were used. The primary platform was a 15-inch Hexsoon EDU-650 quadricopter, providing a stable testbed for varying payload configurations (Figure 2) with a maximum flight time of 30 minutes. Additionally, an agile and smaller 7-inch First Person View (FPV) drone was used in specific locations to evaluate performance at higher velocities and under tighter maneuverability requirements. Both platforms are based on Ardupilot and were equipped with GPS modules accepting Real-Time Kinematic (RTK) positioning corrections to increase the spatial accuracy of the captured signals. The modular payload weighs approximately 300 g, making it easily adaptable to any UAV with a similar payload capacity.

B. PAYLOAD AND DETECTION HARDWARE

The payload includes a Bluetooth receiver, an LNA, an antenna, and a long-range communication system. The actual sensitivity level in BLE is defined as the receiver input signal level for which the Bit Error Rate (BER) is equal to 10^{-3}. While common manufacturers (e.g., Texas Instruments or Espressif Systems) offer sensitivities around -94 dBm, we utilized the nRF52840-DK [38]. This development kit was selected for its slightly superior sensitivity (-95 dBm) and its hardware flexibility, featuring multiple General-Purpose Input/Output (GPIO) ports that we use for data logging on an SD card and a wide power input range (1.7 to 5 V), which allows for direct integration with the drone's flight controller.

The data flow within the UAV is designed for real-time telemetry integration. Once the nRF52840-DK identifies a valid BLE advertising packet, the relevant payload data is encapsulated and sent via a UART interface to the UAV's flight controller. The flight controller then injects this information into the existing LoRa-based telemetry link, effectively acting as a bridge that transmits the device identifier to the GCS without requiring a separate radio link for the BLE data.

The scalability of the proposed system in device-dense environments is primarily defined by the hardware limits of the scanner and the throughput of the telemetry link. The nRF52840 sniffer experiences a processing *blind time* of roughly 50 μs after each successful detection during which it cannot capture new packets. Thus, it can handle channel occupancies of up to 40% (equivalent to approximately 950–1000 packets per second) [39]. Although this capacity is more than sufficient for remote SAR operations, the low-bandwidth LoRa bridge still constitutes a potential bottleneck if every raw packet were to be forwarded. To mitigate this, the system implements firmware-level filtering based on Manufacturer ID and payload size. For scenarios with higher device density, a temporal deduplication procedure should be employed where unique MAC addresses are reported only once within a specific time window (e.g., every 10 seconds). This strategy ensures that the GCS receives critical presence data without saturating the telemetry buffers, even when the 2.4 GHz scanner is operating under high load.

To significantly extend the system's detection range, a key limitation identified in Section VI-A, we integrated a QPL9057 LNA [40]. This component provides a gain of 21 dB with a low noise figure of 0.42 dB, enabling the detection of weak advertising packets that would otherwise be lost, especially in areas with high-density vegetation.

The impact of this integration on the overall system sensitivity can be quantified using Friis' formula for the noise factor of cascaded stages adding the QPL9057 LNA as the first stage. The total noise factor $FTotal$ is determined as:

$$F_{Total} = F_{LNA} + \frac{F_{DK} - 1}{G_{LNA}} \tag{1}$$

The internal noise figure of the standalone nRF52840-DK (NF_{DK}) is not directly provided in its datasheet. However, it can be derived from the performance data of the nRF21540 range extender [41], which claims to improve the nRF52840 SoC sensitivity from -95 to -100 dBm. Applying Friis' formula again, but considering in this case the nRF21540 as the first stage, with gain (13 dB) and noise figure (2.7 dB) allows the estimation of NFDK to 8.33 dB.

Then, the resulting system noise figure using the QPL9057 LNA is 0.60 dB. This represents a reduction of 7.73 dB in the noise floor compared to the standalone development kit. Consequently, the theoretical sensitivity of the system is enhanced from -95 dBm to -102.73 dBm, a significant improvement that directly correlates with the increased detection range observed in our field tests.

C. ANTENNA SELECTION

The radiation pattern and gain of the antenna are decisive factors in the effectiveness of the search mission. In our study, three different antenna configurations were evaluated to identify the optimal balance between coverage area, weight, and signal sensitivity:

- **ImmersionRC SpiroNet 2.4 GHz Patch**: A circularly polarized (RHCP) directional antenna. It provides a high gain of approximately 13 dBi with a half-power beamwidth around 35°. While its RF performance is excellent, its weight of approximately 300 g represents a significant mechanical load and increases aerodynamic drag for small UAVs.
- **Huber+Suhner Sencity Spot-S**: Another professional-grade industrial antenna. It offers a gain of 8.5 dBi with vertical (linear) polarization and a beamwidth of 75°. Despite being lighter (110 g), its robust construction, technical specifications, and its linear polarization make it more sensitive to the UAV's orientation relative to the target.
- **Lynx 2.4-CPA antenna**: A custom-built antenna specifically designed for this project. Although its design results in a slightly lower gain, remaining below 5 dBi, this is intentionally balanced by its superior 110° beamwidth and its circular polarization. Being the most lightweight option with just 27 g, it is ideally suited for the discovery phase of SAR missions, as it maximizes

TABLE 2. Comparison of antenna characteristics.

	ImmersionRC	Huber	Lynx
Beamwidth	35°	75°	110°
Weight	∼300 g	∼110 g	∼27 g
Polarization	RHCP	Linear	RHCP
Gain	13 dBi	8.5 dBi	4.5 dBi
Size (mm)	200x200x15	101x81x36	40.5x40.5x28

the instantaneous search area while minimizing the impact on the UAV's flight dynamics and endurance.

Table 2 summarizes the trade-offs observed. The Lynx antenna was prioritized for the initial discovery phase of the SAR missions. Its wider aperture allows the UAV to sweep a larger ground area in a single pass, effectively increasing the search area rate. Furthermore, its lightweight construction, optimized for 3D-printed mounts, minimizes aerodynamic drag and mechanical stress on the drone's gimbal compared to the heavier commercial alternatives.

D. GCS AND CLOUD INTEGRATION

The GCS serves as the termination point for the long-range telemetry link. Physically, the GCS consists of a standard laptop interfaced with a LoRa receiver. This receiver captures the UAV's telemetry information including the packets containing the decoded BLE information from the UAV.

For integration into global tracking ecosystems, we propose a software-based bridge. The captured telemetry data can be processed using a Python script to extract the original BLE advertising payload. This payload would then be sent to a secondary nRF52840-DK located at the GCS, which acts as a "re-emitter". By placing a cellular-connected Apple or Google device in close proximity to this re-emitter, the system can spoof the device's presence at the GCS's coordinates, thereby triggering a location update on the *Find My* or *Find My Device* networks and confirming the SAR command center in real-time whether the detected device was the targeted one.

E. IMPLEMENTATION DETAILS AND RELAY WORKFLOW

To ensure reproducibility, the end-to-end relay architecture is detailed as follows. At the airborne node, the nRF52840 sniffer filters packets by Manufacturer ID and payload size at the firmware level. This sniffer is implemented using the Nordic nRF Connect SDK v3.2.1 based on Zephyr RTOS.

Validated packets are forwarded via UART to the flight controller, where a simple Lua script captures the data and injects it into the LoRa-based telemetry stream.

At the GCS, the processing is split between two specialized Python scripts because the Python interpreter inside Mission Planner is quite limited. In this way, first, a simple script running within Mission Planner environment captures the telemetry and logs the BLE payloads to a local file. On the other hand, simultaneously, another Python script running directly on the operating system monitors this file in real-time to extract and forward the data to the re-emitter.

The re-emission is handled by a secondary nRF52840-DK programmed with a basic implementation of Zephyr including the *BT* and *SHELL* modules necessary to expose an HCI interface over UART. This allows the GCS to clone the original packet using the following specific HCI command sequence:

1) **Initialization:** The controller is reset and initialized
   `bt disable`
   `bt init`
2) **Advertising MAC address configuration:**
   `bt hci-cmd 08 05 <MAC_Address>`
   (Captured address)
3) **Advertising parameters configuration:**
   `bt hci-cmd 08 06 ...`
   `20032003000101XXXXXXXXXXXX0700`
   (In this case we set a 500 ms advertising interval over the three primary advertising channels. The XX represent the MAC Address previously configured.)
4) **Payload Loading:**
   `bt hci-cmd 08 08 <Captured_Payload>`
5) **Enable advertising:**
   `bt hci-cmd 08 0a 01`

We used a Nordic development kit as the re-emitter, but following this implementation, any other device capable of accepting HCI commands on a command-line could be used.

Final validation in the Find My network was conducted manually. When a location update is observed in the Find My application (triggered by the GCS's proximity to a *finder* device), the event is manually timestamped to cross-reference the exact drone coordinates with the cloned packet, confirming a successful identification in a communication-deprived environment.

V. EXPERIMENTAL SETUP AND FIELD TESTS

The validation of the proposed system was conducted through extensive field tests across three diverse geographical locations in Spain: Begues (peri-urban forest, Fig.3), Rasos de Peguera (high-altitude variable terrain, Fig.4), and Barranco Hondo (extreme canyon environment, Fig.7) to evaluate the system's performance across various topographies while strictly adhering to European drone regulations.

All operations were performed under the Open Category A3. This framework mandates that flights occur in areas where no uninvolved persons are present and at least 150 meters away from residential or recreational zones. Operationally, this requires maintaining Visual Line of Sight (VLOS), which effectively limits the maximum distance from the pilot to 500 meters. All tests were conducted below 120 meters Above Ground Level (AGL), with a constant UAV flight speed of 5 m/s and antennas oriented in a nadir position.

For all scenarios, we used several control devices based on Nordic nRF52840 development kits. These were configured to broadcast generic on/off BLE mesh messages (39-byte beacons) with an advertising interval of 250 ms and a transmission power of 4 dBm, which is in the order of magnitude of the devices described in section III-B (the PCB antenna of the nRF52840-DK has approximately 0 dBi [42]).

To simulate realistic search conditions, the devices were placed in closed plastic boxes on the ground, often hidden near trees or rocks (waypoints A, B, C, and D), away from trails, and ensuring RF isolation between them at ground level. Waypoint G in figures marks the location of the GCS.

A. TEST SCENARIO 1: BASELINE TESTING IN A PERI-URBAN FOREST

This initial scenario (Figure 3) was conducted in a small wooded area on the outskirts of Begues, near the university campus. Due to its proximity to the Barcelona-El Prat airport, flight altitude was strictly capped at 50 meters AGL which provided sufficient clearance to overfly the 20-25 m tall trees. This scenario served to establish a technical baseline for the three antenna types (Lynx, ImmersionRC, and Huber+Suhner) to evaluate initial ground footprint and link reliability in a constrained environment.

B. TEST SCENARIO 2: HIGH-MOUNTAIN SEARCH USING A LNA

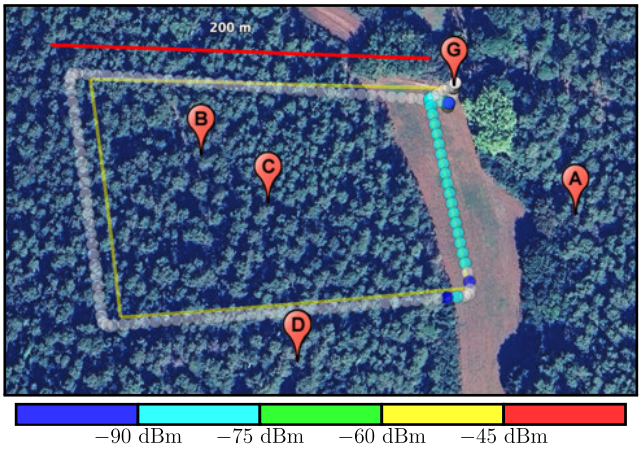
The second location was a decommissioned ski resort in Rasos de Peguera at an altitude of approximately 2000 meters. This isolated environment, surrounded by dense 30 m tall pine forests, provided a radio-quiet setting where the only significant interference in the 2.4 GHz band came from the personal devices (smartphones, laptops, and smartwatches) of the research team. This location allowed for more ambitious flight plans, both in altitude and distance, reaching near the regulated limits.

Here we introduced the LNA to enhance the results over this larger search area, and executed two different flight plans:

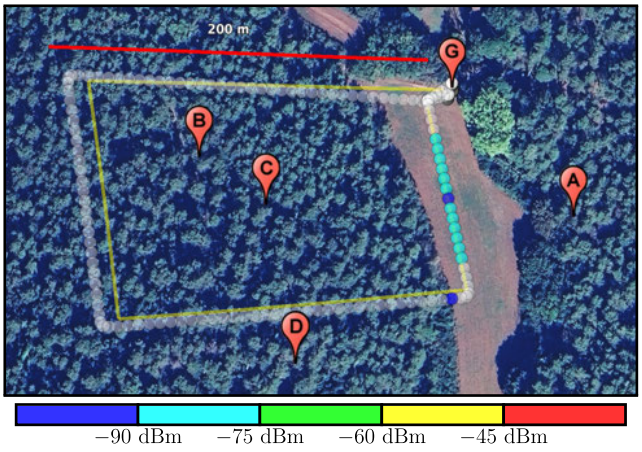
- Hotspot spiral: A square area with a spiral sweep flight path (25 m between segments) to evaluate local detection density. Figure 4a, illustrates this flight at 100m AGL using the Lynx antenna and detecting Device B transmissions. In this scenario, all three antennas were tested at altitudes of 50 m and 100 m AGL.
- Long-range rectangular: A trajectory designed to test maximum range within A3 category limits. The path is depicted in Figure 4b with a longer side of 450 m and 50 m before turning around into the next sweep.

C. TEST SCENARIO 3: SLOT CANYON

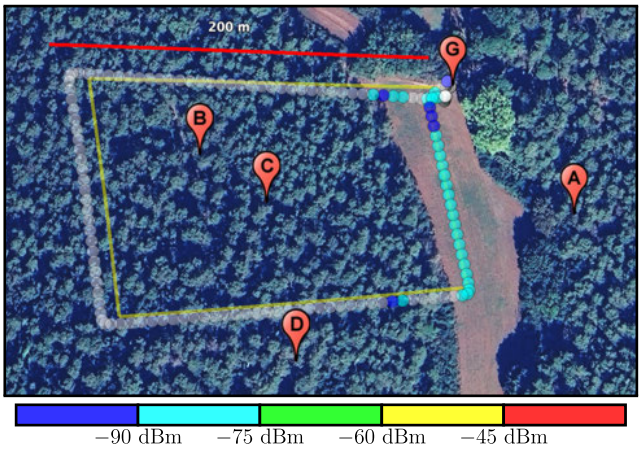
The final test took place in the Guadalaviar river basin, specifically in a slot canyon environment characterized by extreme narrowness and vertical rock walls with depths around 40 m. This location represents one of the most challenging scenarios for SAR operations, as the canyon is so narrow in certain sections that a person can touch both walls simultaneously by extending their arms (see Figure 7a). In this case only two control devices were placed, both on the riverbed, in areas with no cellular coverage and zero GPS signal reception due to the canyon's winding topography. Again, the devices were positioned such that they were mutually RF isolated. We utilized this unique environment to perform a direct comparative study on the impact of the LNA.



(a) Huber-Suhner antenna



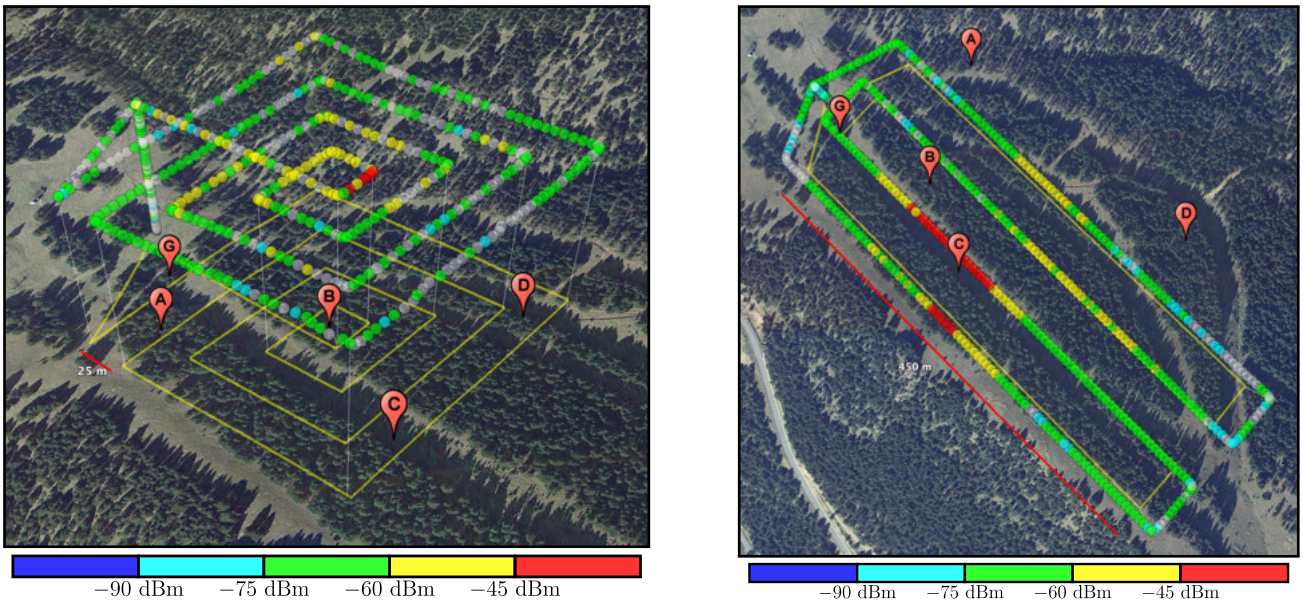
(b) ImmersionRC antenna



(c) Lynx antenna

FIGURE 3. Device A detections in a peri-urban forest.

Although the control devices experienced no GPS signal reception, the UAV, flying above the rim, maintained a high-quality GNSS fix with 14 to 16 satellites. As RTK positioning corrections were used, the achieved horizontal positional uncertainty was of approximately 0.01–0.03 m



(a) Hotspot flight plan: Device B detections with ImmersionRC antenna at 100 m

(b) Long range flight plan: Device C detections with ImmersionRC antenna at 100 m

FIGURE 4. Conducted flight plans for the high-mountain scenario.

throughout all the flights. This high level of precision ensures that the coordinates associated with each BLE packet capture accurately reflect the drone's location at the time of detection, which is critical for the subsequent localization of the victim.

The UAV performed identical flight paths over the canyon at a constant altitude, once with the LNA integrated into the payload and once without it, both with the Lynx antenna. All other variables (flight speed, and target positions) remained constant.

Additionally, this scenario was used to validate the end-to-end relay functionality of a packet emitted by a commercial AirTag, simulating a real-world lost-item or victim scenario in areas without cellular network coverage.

VI. RESULTS AND DISCUSSION

In this section, the data captured during the field tests are analyzed, correlating the Received Signal Strength Indicator (RSSI) values with the slant path and evaluating the impact of hardware design choices on search efficiency.

A. SCENARIO 1: BASELINE AND INITIAL DETECTION IN A PERI-URBAN FOREST

The first experimental campaign in Begues served to establish the system's baseline performance. As previously described, this scenario was executed without the use of the LNA and was restricted to a flight altitude of 50 m AGL due to the proximity of Barcelona-El Prat Airport.

The figures corresponding to this scenario (Figure 3) display the results for the most isolated control node (Device A) using the three different antenna types. The colored points represent the average RSSI level of the received packets, acquired during one second, according to the scale shown in the color bar, while the translucent white points indicate UAV positions where no packets were received from that specific device. The yellow line represents the ground projection of the flight path.

Although Device A is detected by all three antennas, a comparison of the three subfigures shows that the Lynx antenna presents a better footprint; with more detections and range than the other two more directive antennas, providing the first practical evidence of how a wider beamwidth (110°) can maximize the probability of detection during a single flight pass. Furthermore, the data reveals that the UAV must maintain significant proximity to the target, as consistent detections only occur when the slant path remains approximately below 100 m. Therefore, while the BLE signal is inherently detectable, its operational range seems insufficient for the requirements of wide-area SAR missions. This fact served as the primary technical justification for integrating an LNA in the subsequent and more demanding experimental scenarios.

B. SCENARIO 2: HIGH-MOUNTAIN SEARCH AND ALTITUDE ANALYSIS

The second campaign allowed for a more comprehensive evaluation of the system, now equipped with an LNA, under varying altitudes and flight patterns. This environment, characterized by low electromagnetic interference, was ideal for testing the maximum detection range and the behavior of the different antennas at 50 m and 100 m AGL.

The system's behavior during dense search patterns is depicted in Figure 4a, which shows the Hotspot spiral flight at 100 m AGL. In this specific test, the Lynx antenna was used to detect transmissions from Device B. A crucial observation

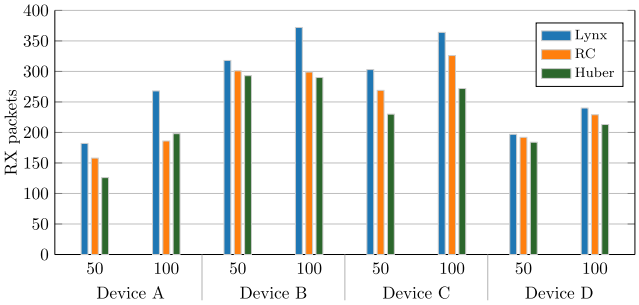


FIGURE 5. Received packets per device and antenna at different altitudes (m) for the hotspot flight plan.

in this scenario is that, with the LNA integrated, the device is detected almost continuously throughout the entire flight duration. The visualization shows a high density of colored points with consistently high RSSI values centered around the target, confirming that the combination of the amplification and the wide-beam antenna creates a robust detection zone that remains effective even as the UAV banks during the tight turns of the spiral.

The total packet count for each configuration of antenna, flight altitude and detected device is summarized in Figure 5, which focuses on the data obtained during the Hotspot spiral trajectory (Figure 4a). The data clearly indicates that the Lynx antenna consistently outperforms the commercial alternatives in terms of raw packet reception, regardless of the flight altitude. However, the impact of increasing the altitude to 100 m was particularly beneficial for the ImmersionRC antenna; its narrower beamwidth covers a significantly larger ground footprint area at higher altitudes, leading to a proportional increase in detections. Conversely, the Huber-Suhner antenna exhibited the poorest performance due to its linear polarization, which caused significant packet loss during the UAV's spiral maneuvers as the polarization axes became misaligned with respect to the target device.

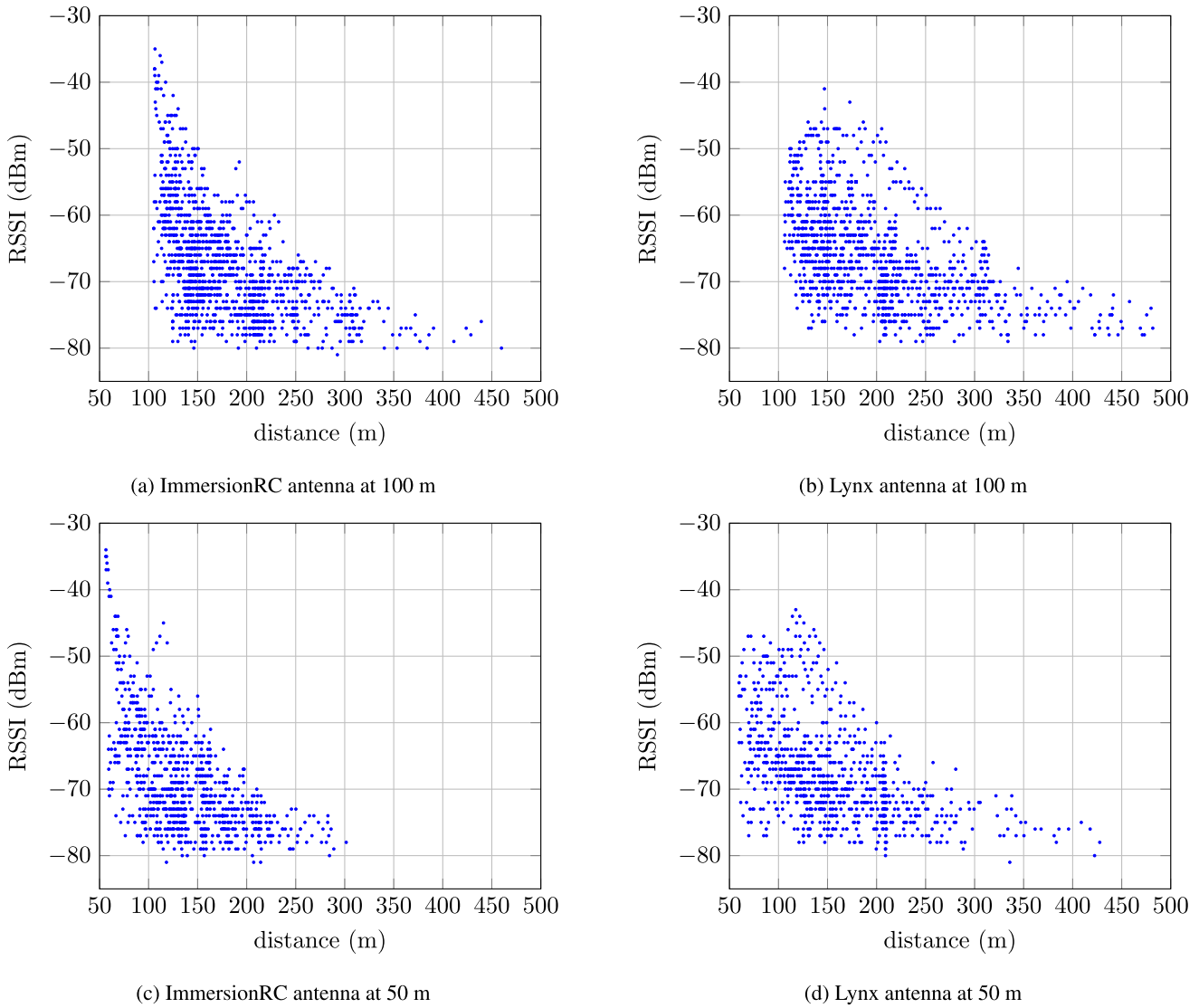
A localized reduction in packet counts can be noted for Device A. This was primarily due to its proximity to the GCS in that specific flight plan. The personal electronic devices (smartphones, laptops, and wearables) used by the research team generated a more saturated radio environment; as the scanner processed numerous advertising packets from these unintentional sources, it led to the dropping or collision of packets from the intended control nodes. This effect was reduced in the long-range rectangular flight, as the device was relocated further away from the GCS for that experiment. For maximum range evaluation, the long-range flight plan was executed both for the Lynx and the ImmersionRC antennas. A representative example of this trajectory is shown in Figure 4b, which shows the detections for Device C using the ImmersionRC antenna at 100 m AGL. Notably, when flying directly over the selected device, the ImmersionRC antenna's higher gain produces a significant concentration of red points, which facilitates precise localization. Furthermore, the data shows that by incorporating the LNA and increasing the altitude to 100 m, even this more directional antenna manages to capture packets during nearly the entire path, resulting in very few translucent white points compared to the baseline tests.

A detailed analysis of the signal behavior is provided by the scatter plots in Figure 6, which depict the total logged measurements of the four control devices during this long-range flight against the slant range between the devices and the drone. Analyzing these subfigures by vertical pairs allows for an assessment of the effect of flight altitude for the same antenna. In both cases, as the altitude increases, the plot shifts to the right and the point density increases, indicating a higher number of detections and a greater overall reach. This effect is especially pronounced for the ImmersionRC antenna, where the gain in altitude significantly expands its detection capability.

Comparing the subfigures horizontally reveals the effect of the chosen antenna for the same flight altitude. Beyond a 200 m distance (or 250 m when flying at 100 m AGL), the Lynx antenna exhibits a much higher volume of points than the ImmersionRC. However, the dispersion of the point cloud is generally greater for the Lynx due to its wider beamwidth and the influence of vegetation. This suggests that while we can detect a device further away, the Lynx antenna acts more as a "presence detector" than a precision localizer at those extreme ranges. For instance, at 100 m AGL, an RSSI reading of -50 dBm with the Lynx antenna might indicate the device is anywhere between 110 and 210 m away. In contrast, the ImmersionRC antenna offers much higher precision for that same power level, narrowing the distance estimate to between 110 and 150 m. This precision is reflected in the sharp peak of values on the left side of the ImmersionRC plots; receiving a very high power value indicates that the UAV is overflying the target, a fact corroborated by Figure 4b, where points exceeding -45 dBm are marked in red.

Ultimately, analyzing all four subfigures confirms the maximum detection slant paths achieved. As expected, the Lynx antenna provides the highest values thanks to its 110° beamwidth. Specifically, the system successfully detected Device A at a distance of 428 m at 50 m altitude, reaching a maximum of 482 meters when flying at 100 m. This level of performance, enhanced by the LNA, ensures a robust link that remains active for nearly 100% of the mission time. From another point of view, this footprint allows for a high search area rate. Considering the 5 m/s search velocity, a straight flight, and a conservative coverage of 300 m, the effective surveyed area would exceed 900 000 m^2 (222 acres) in just five minutes, significantly reducing the time-to-discovery compared to narrower commercial sensors that would require multiple overlapping passes to achieve the same ground coverage.

To contextualize the technical advancements of this work, our performance metrics are compared with the core studies that established the feasibility of airborne BLE sensing and terrestrial RF benchmarks in mountain rescue. Initial studies by Hashmi [13] and Cantizani-Estepa et al. [12] utilized standard hardware tools, such as the Raspberry Pi 3 and ESP32 SoCs, which are limited by the base sensitivity



(a) ImmersionRC antenna at 100 m

(b) Lynx antenna at 100 m

(c) ImmersionRC antenna at 50 m

(d) Lynx antenna at 50 m

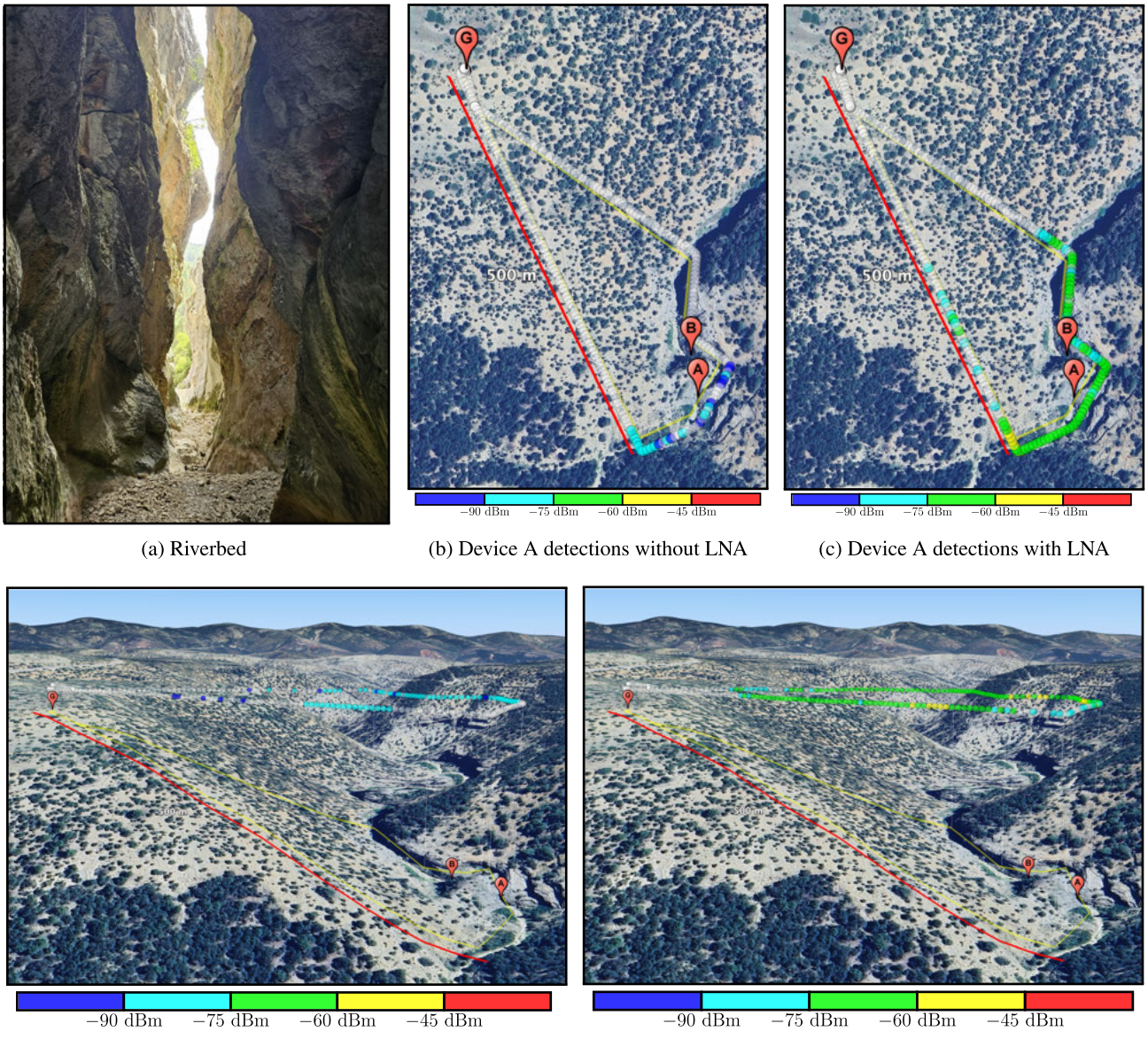
FIGURE 6. Scatter plots of all received packets from devices A, B, C and D in the long-range scenario.

of their integrated Bluetooth controllers (typically around -94 dBm). Consequently, their findings were restricted to short-range proximity detection: Hashmi defined a maximum coverage radius of 10 m for indoor planning [13], while Cantizani-Estepa reported reliable detection up to 15 m in forest environments [12]. Moving toward mountain-specific benchmarks, terrestrial evaluations by Bianco et al. [24] using LoPy-4 programmable boards established that standard Bluetooth (v4.2) and Wi-Fi (802.11n) achieve maximum line-of-sight ranges of 192 m and 236 m, respectively. In contrast, our system—leveraging an airborne perspective, the QPL9057 LNA, and custom Lynx antennas—achieved a maximum slant range of 482 m, representing a 30-fold increase over robotic sniffers and doubling the terrestrial BLE baseline. This range extension directly correlates with superior search efficiency. While the system in [13] reported surveying a 3,600 m^2 area in five minutes (720 m^2/min), our LNA-enhanced UAV demonstrated an effective coverage of 900,000 m^2 in the same time frame (180,000 m^2/min). Long-range technologies like LoRa have been empirically proven by Bianco et al. [24] to maintain links up to 8.4 km in mountainous terrain. However, such systems require the victim to carry a specific pre-registered LoRa radio.

C. SCENARIO 3: NARROW CANYON EXPLORATION

The third experimental campaign in Teruel was designed to test the system's performance in one of the most hostile topographies for radio frequency propagation: a narrow limestone canyon ("Barranco Hondo") characterized by vertical walls and riparian vegetation (Figure 7a). In this environment, the objective was to analyze how the system handles radio shadows and extreme signal diffraction.

During the flights at over 100 m AGL relative to the canyon floor, the data revealed that the LNA is a critical requirement for operating in such complex geometries. As shown in Figures 7b and 7d, without amplification, the rock walls



(a) Riverbed (b) Device A detections without LNA (c) Device A detections with LNA

(d) Device B detections without LNA (lateral view) (e) Device B detections with LNA (lateral view)

FIGURE 7. Detections in a slot canyon.

of the canyon effectively masked the BLE signals for the majority of the flight path for both devices. However, with the 21 dB gain provided by the LNA and its low-noise figure, the system was able to capture packets even when the UAV was not in a direct line-of-sight with the devices at the bottom (Figures 7c and 7e), leveraging signal diffraction over the canyon edges. The Lynx antenna's wide 110º beamwidth proved advantageous here, as it allowed for a larger detection window compared to the narrower commercial patches, which would have required a much more precise overflight of the canyon axis to obtain a similar number of detections.

The resulting RSSI values in the canyon, visualized in the path reconstruction of Figures 7c and 7e, were generally lower compared to the open-field tests in Rasos de Peguera, a direct consequence of the multipath fading and signal absorption by the rock and vegetation. Nevertheless, the continuous detection of control devices at the canyon bed demonstrates that the system remains operationally effective even in locations where GNSS and cellular signals are severely degraded or absent. These results confirm that the combination of a high-sensitivity scanner and a wide-beamwidth circular polarization antenna is capable of overcoming the slot canyon effect, providing SAR teams with a reliable tool for exploring areas that are otherwise inaccessible without significant ground effort.

D. SLANT RANGE AND PACKET DELIVERY ANALYSIS

All results and conclusions presented in this manuscript were derived from 76 systematic detection attempts

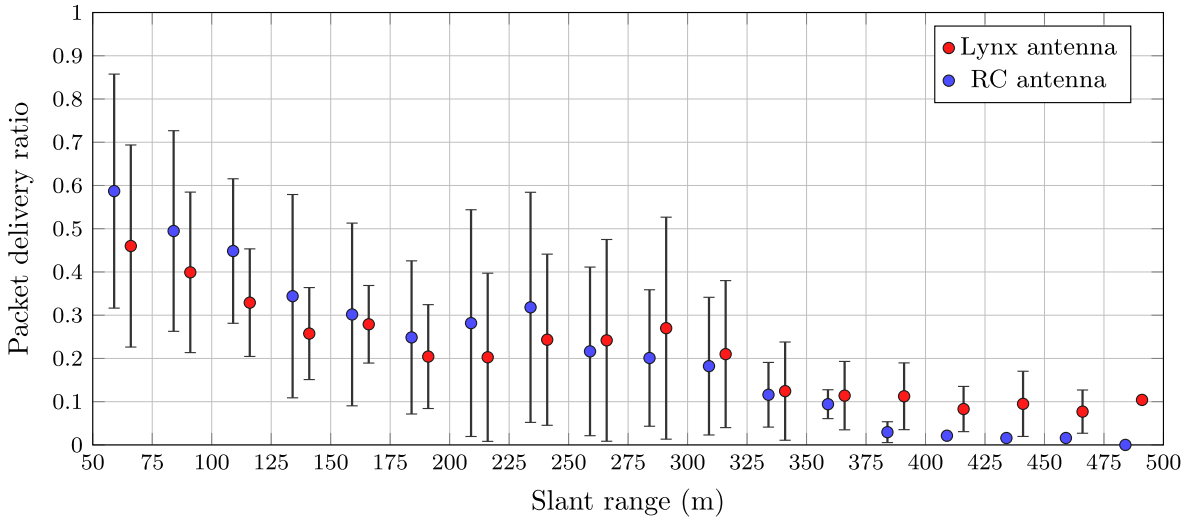


FIGURE 8. Packet delivery ratio vs. slant range for the Lynx and ImmersionRC antennas.

conducted across diverse geographical scenarios and hardware configurations. Throughout these field trials, the system demonstrated exceptional reliability, achieving a 100% device detection rate, with packet captures ranging from a minimum of 38 to a maximum of 1148. Given that the proposed BLE-to-LoRa relay requires only a single valid packet to trigger a location update in the cloud, this consistent volume of traffic ensures total operational success.

Nevertheless, to provide a more statistical evaluation of the system's performance, this section analyzes the Packet Delivery Ratio (PDR) and detection probability derived from a subset of the field trials to characterize the underlying link reliability. Figure 8 illustrates the PDR as a function of the slant range using data from 36 specific trials, those equipped with the LNA, including different approximations to the target device, flight plans and flight heights using the two final selected antennas (Lynx and ImmersionRC). The PDR is estimated by dividing the slant range into 25-meter bins and comparing, on a per-bin basis, the captured packets against the expected transmissions. These transmissions are calculated analyzing the time the drone was within that bin and considering the 250 ms advertising interval of the control devices.

The results, presented with error bars representing one standard deviation, reveal a notable level of dispersion. This variance is technically expected, as slant range alone does not account for orientation and/or environment conditions. For instance, the PDR at 150 meters varies greatly depending on whether the drone is approaching or moving away from the target, as the antenna orientation shifts significantly during flight maneuvers. Similarly, environmental shadowing differs by altitude, where a 50-meter flight path involves more signal penetration through tree canopies than a 100-meter flight, which probably offers a clearer line-of-sight at identical ranges.

The quantitative data confirms again that the ImmersionRC antenna offers superior performance at close proximity (under 200 m) due to its 13 dBi gain. However, the custom Lynx antenna proves more effective for wide-area discovery, maintaining a higher and more stable PDR at ranges exceeding 250 m.

It is important to emphasize that while the instantaneous PDR may seem very low with distance, the overall detection probability (P_d) is substantially higher because it is a cumulative function of the time the UAV remains within range, defined as $P_d = 1 - (1 - \text{PDR})^N$, where N is the total number of packets transmitted. For example, if a drone maintains its flight path at a slant range between 325 and 500 m for 30 seconds, an AirTag (transmitting once every two seconds) would send 15 packets. Even with a 10% PDR observed at these distances, the calculated detection probability would be 79.4%.

Empirically, this cumulative effect explains why our 76 systematic field trials consistently achieved a 100% success rate, as the time-of-flight over the target areas always allowed for a sufficient number of transmissions to guarantee at least one successful capture. Even in the worst-case scenario, characterized by low altitude (50 m), a directive antenna, no LNA, and with the target outside the flight perimeter, the system captured 38 packets. This result is significant because it indicates that while only a single packet is required for a detection, the device actually transmitted a much higher volume of advertisements during that window, ensuring a wide margin of success. Consequently, if a systematic grid flight plan is implemented such that the search area is constrained within a perimeter where all points are guaranteed to be within the system's operational reach, the overall detection probability can be further elevated, providing a highly robust tool for professional SAR teams in critical environments.

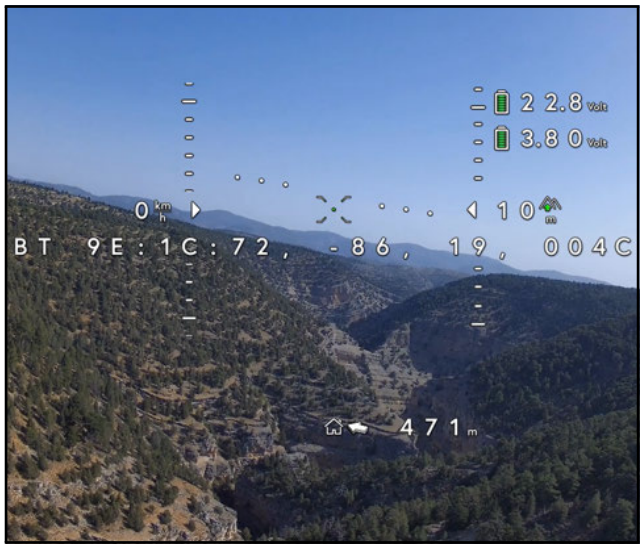


FIGURE 9. AirTag detection from the pilot point of view.

E. REAL-TIME DEVICE IDENTIFICATION (BLE-LORA-CLOUD RELAY)

To conclude, another critical challenge in BLE-based search operations is the identification of the target. Since MAC addresses are often randomized or unknown to the rescue teams, detecting a generic Bluetooth packet does not guarantee it belongs to the person being sought. To address this, the system depicted in Figure 1 incorporates a specialized validation workflow tested with Apple AirTags, but also applies to similar crowdsourced network devices. By filtering incoming traffic based on a specific Manufacturer ID and payload size, the airborne nRF52840-DK sniffer identifies potential AirTag advertisements in real-time.

When such a packet is identified, it is encapsulated and transmitted via the LoRa link to the GCS. At the base station, a Python-based script processes the incoming LoRa data and forwards it to a secondary nRF52840 development kit. This device is configured as a standard BLE peripheral that accepts HCI (Host Controller Interface) commands via a virtual serial port (USB). This setup allows the GCS to clone the exact captured advertising packet into the local environment. With an internet-connected iOS device (iPhone or iPad) present near the GCS, it detects this re-emitted packet and automatically updates the location in the Apple "Find My" cloud. This process effectively bridges the gap from an isolated area to the global search infrastructure, uniquely identifying the victim.

It should be noted that the location registered in the cloud corresponds to the GCS coordinates, rather than the actual position of the victim. Therefore, once the device is identified, the search team relies on the logged UAV position associated with the original packet capture to pinpoint the target's locations in the field.

This operational capability is illustrated in Figure 9, which shows a capture of the pilot's Head-Up Display (HUD) during a validation flight also in third scenario. In this test, conducted at a speed of 7.5 m/s with the detection system (Lynx+LNA+sniffer) loaded on the 7-inch FPV drone and following the previous path, the HUD displays a real-time detection overlay. The center of the screen shows the critical parameters of the captured packet: the last three bytes of the Bluetooth MAC address (BT 9E:1C:72), the current RSSI (−86 dBm), the hexadecimal payload size (0 × 19), and the Manufacturer ID (0 × 004C). This visual feedback allows the pilot to confirm target acquisition in situ, even before the cloud update is processed, demonstrating a robust end-to-end integration for professional SAR workflows.

VII. CONCLUSION

The experimental results presented in this study validate the feasibility of using UAV-mounted Bluetooth scanners as RF remote sensors for high-efficiency SAR operations across diverse and challenging topographies. The primary findings underscore that the integration of a LNA is a fundamental necessity rather than a mere enhancement for overcoming signal diffraction and extreme path loss in complex environments such as slot canyons and dense forests. Without the LNA, the detection was limited to a nadir-only geometry, requiring the UAV to be almost perfectly aligned with the target. With the LNA, the detection cone expanded, which in a real mission translates to a higher tolerance for flight path errors and a reduced risk of missing a victim hidden under dense canopy or rock formations.

Furthermore, the performance of the custom-designed Lynx antenna demonstrated that a wide beamwidth of 110° is strategically superior for the discovery phase of SAR missions, providing a significantly larger ground footprint that maximizes the probability of detection during high-speed aerial sweeps compared to high-gain but narrow-beam commercial alternatives. To achieve the same area coverage as one Lynx pass (900 000 m^2 in five minutes), a UAV equipped with a narrow-beam antenna would need to perform multiple parallel sweeps, significantly increasing the battery consumption and time-to-discovery. In contrast, when using more directive antennas, our tests confirm that increasing the flight altitude is highly recommended to compensate for the reduced ground coverage and ensure adequate overlap between flight lines. Additionally, circular polarized antennas in the aerial sniffer are recommended to mitigate the effects of UAV banking and orientation changes.

The operational effectiveness of the system is further supported by the achievement of a maximum slant path of 482 meters at a flight altitude of 100 meters. This reach aligns precisely with the 500-meter VLOS limit established by European A3 Open Category regulations. While real-world SAR operations are often exempt from certain civil aviation restrictions during emergencies, our strict adherence to the A3 category during these trials confirms that the hardware architecture is already optimized for operation within standard legal and safety frameworks if the search is not conducted by an emergency team.

Finally, the successful validation of the Bluetooth-LoRa-Cloud bridge confirms the system's ability to reconnect isolated victims in environments without network coverage

with global tracking infrastructures like Apple's Find My network. This turns any standard AirTag or BLE-enabled device into a long-range emergency beacon without requiring hardware modifications on the victim's side, offering a low-cost and highly scalable solution that leverages the ubiquity of personal Bluetooth devices to significantly reduce time-to-discovery in life-critical scenarios.

Future work will focus on evaluating performance with a broader set of commercial BLE devices in real SAR exercises and extending the approach to multi-UAV coordination. In this case some of the drones would carry wider-beamwidth antennas for quick target detection and others would be loaded with more directive antennas for precise location. The latter would deviate from the original flight plan to pinpoint the target, integrating automated flight-path optimization based on terrain and prior success probabilities increasing even more the utility of UAVs in SAR missions.

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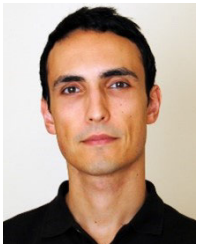


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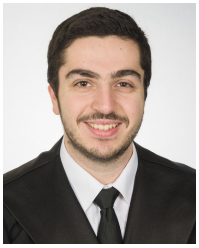


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